

An Asynchronous Energy-Harvesting Pixel for CMOS Image Sensors

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Abstract—This letter presents an asynchronous energy-harvesting active pixel that provides a novel solution for self-powered image sensors. The proposed design employs a negative-voltage switch that automatically transitions the photodiode from imaging mode to energy-harvesting mode when the photodiode voltage drops below a threshold, without requiring synchronized timing control. By utilizing idle windows within the frame period, the pixel can achieve simultaneous imaging and energy harvesting within a single frame cycle. The prototype, fabricated in a standard 0.18- μm CMOS process, operates at a 1 V supply with a frame rate of 35.9 fps, consuming a total power of 3.98 μW . It achieves a dynamic range of 45 dB, a pixel fill factor of 55.76%, and an energy-harvesting efficiency of 60.1 pW/lux/mm² while maintaining full imaging functionality.

Index Terms—asynchronous pixel, image sensor, energy harvesting.

I. INTRODUCTION

SELF-powered image sensors play an irreplaceable role in environmental monitoring [1], [2] especially in remote or hazardous areas where traditional battery-powered solutions are impractical [3]. They offer distinct advantages including ultra-low power consumption, simplified deployment, and significantly reduced maintenance costs.

Energy harvesting technology is the core of self-powered image sensing [4]. However, existing technologies predominantly adopt time-division multiplexing approaches [5]–[7]. This technique requires configuring photodiodes into either imaging mode or energy harvesting mode at different times under unified timing control, while relying on off-chip energy storage devices to achieve self-powering. Consequently, the sensor cannot maintain continuous operation. In contrast, the time-to-first-spike (TFS) pixel technology enables the pixels to autonomously switch to energy harvesting mode after imaging readout [8], [9]. This architecture eliminates the need for unified timing control through an event-driven mechanism,

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allowing the pixels receiving strong light to prioritize energy harvesting. Compared to conventional time-division multiplexing pixels, it offers extended energy harvesting duration. However, its event-stream output format necessitates dedicated decoding circuitry, increasing system hardware complexity and costs. More critically, the intricate pulse-generation circuits and timing control units significantly reduce the photosensitive area, resulting in a fill factor typically below 40%. This not only severely limits spatial resolution but also diminishes energy harvesting efficiency.

To address these limitations, this letter proposes an asynchronous pixel with integrated energy harvesting capability. The key innovation lies in its ability to concurrently perform energy harvesting and imaging functions within a single frame period, while maintaining full compatibility with conventional APS imaging characteristics. Through optimized pixel structure design, the proposed solution achieves a remarkable 58.37% fill factor. The letter is structured as follows: Section II discusses the circuit structure and operating principles. Section III provides the measurement results, and Section IV concludes the letter.

II. DESIGN AND OPERATION OF THE PROPOSED PIXEL

Fig. 1(a) illustrates the proposed asynchronous energy harvesting pixel circuit. The pixel comprises two functional blocks: the wide-swing pixel circuit and the negative-voltage switch circuit. The wide-swing pixel adopts a combined structure consisting of a PMOS reset transistor, complementary transfer gates, complementary row select transistors, and a source-follower amplifier transistor. This configuration preserves the operating principle of a conventional 4T active pixel while extending the output voltage swing and thereby improving the dynamic range [10]. Since the power consumption of the pixel array accounts for only a minor portion of the total sensor power, the additional dynamic power introduced by the complementary switch has a negligible impact on the overall power consumption. Following image signal readout, the photodiode voltage V_{PD} progressively decays to negative potentials under sustained illumination. To enable dual-mode operation, that is, imaging at positive V_{PD} and photovoltaic energy harvesting at negative V_{PD} , a switching mechanism must selectively connect to V_{EH} to V_{PD} when $V_{PD} < 0$ while remaining isolated when $V_{PD} > 0$. However, conventional switches prove inadequate since its one terminal connects to V_{EH} (the energy harvesting output node typically biased between -400 mV to -300 mV). This work

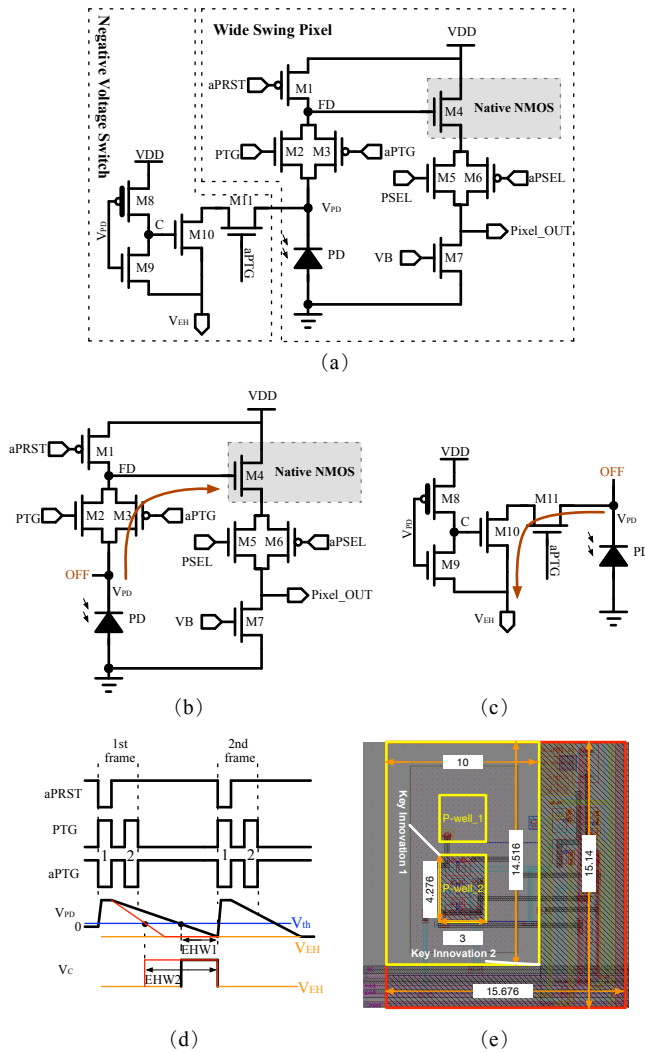


Fig. 1. The proposed asynchronous energy-harvesting pixel: (a) circuit diagram, (b) imaging mode, (c) energy-harvesting mode, (d) timing diagram, (e) the layout.

therefore introduces the novel negative-voltage switch circuit. For $V_{PD} \in [V_{EH} + V_{th}, 1, V]$, the $V_{EH}-V_{PD}$ connection is off, and the pixel operates in the imaging mode [Fig. 1(b)]. For $V_{PD} \in [V_{EH}, V_{EH} + V_{th}]$, the connection is on, and the pixel switches to the energy-harvesting mode [Fig. 1(c)], where V_{th} (≈ 400 , mV) represents the threshold voltage of transistor M9 as annotated in Fig. 1(a). To achieve a switching threshold close to 0 V, the inverter formed by M8 and M9 employs transistors with different voltage ratings (3.3 V for M8 and 1.8 V for M9).

Fig. 1(d) details the operational timing diagram. During initial activation ($aPRST = \text{low}$, $PTG/aPTG = \text{high/low}$), the photodiode PD resets to VDD. Subsequent exposure causes V_{PD} to decay monotonically. At the second $PTG/aPTG$ transition, V_C gets sampled for readout. When V_{PD} drops below M9's threshold, the M8/M9 inverter toggles, pulling node V_C to VDD. With $aPTG = \text{high}$, this activates M10/M11 to initiate energy harvesting until the next frame reset. The V_C timing diagram is shown in Fig 1(d). The portion of the frame period dedicated to energy harvesting can be termed as the Energy Harvesting Window (EHW), with units of second (s).

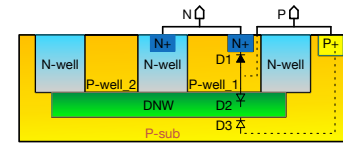


Fig. 2. The proposed photodiode for pixel.

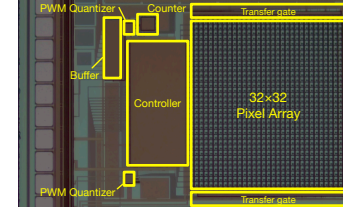


Fig. 3. Micrograph of the prototype chip.

To quantify the contribution of the energy harvesting window, the energy harvesting density per unit light intensity and per unit area can be expressed as:

$$u_{EHW} = \frac{1}{A_{\text{pixel array}} \cdot L} \int_{t_1}^{t_2} P(t) dt \quad (1)$$

where $P(t)$ is the energy harvesting power (which can be measured using a source meter), $A_{\text{pixel array}}$ represents the pixel array area (unit: mm^2), and L denotes the incident light illuminance (unit: lux, measurable via a lux meter). It can be seen that increased illumination accelerates V_{PD} decay, thereby reducing energy harvesting initiation delay (compare EHW1 vs EHW2 durations).

Fig. 1(e) presents the pixel layout, highlighting two key innovations: (1) P-well_2 incorporates deep n-well NMOS transistors M9/M10/M11 with isolated substrates, enabling negative-voltage switching in the digital control path; (2) The strategic coverage of metal over the MOS transistors within the pixel is employed to mitigate the impact of illumination on the device. Consequently, P-well_2 does not participate in photodetection, resulting in a fill factor of:

$$FF = \frac{A_{PD} - A_{P\text{-well}_2}}{A_{\text{pixel}}} = 55.76\% \quad (2)$$

Compared with the asynchronous pixel scheme in [8], this represents a 27.72% improvement, while demonstrating a 22.77% enhancement over conventional active pixels [5].

Fig. 2 shows the photodiode used in the pixel design, which adopts a vertical $N^+/P\text{-well}/DNW/P\text{-sub}$ structure. In this design, the N^+ and DNW are shorted to serve as the cathode of the photodiode, while the P-well and P-sub are shorted to serve as the anode, P-well_2 serves as the substrate for the deep-nwell NMOS transistors within the pixel. Compared to the traditional $N^+/P\text{-sub}$ structure, this design features a larger junction capacitance and full well capacity, which helps to improve the dynamic range [11]. When the photodiode is configured in the photovoltaic mode, the N-side is used as the energy harvesting output. In contrast to the design in [9], which uses the P-well as the energy harvesting output, our design incorporates an additional D3 PN junction for energy harvesting, resulting in a higher energy harvesting efficiency.

The readout circuit adopts a dual-channel time-interleaved PWM quantizer, in which the odd and even pixel columns

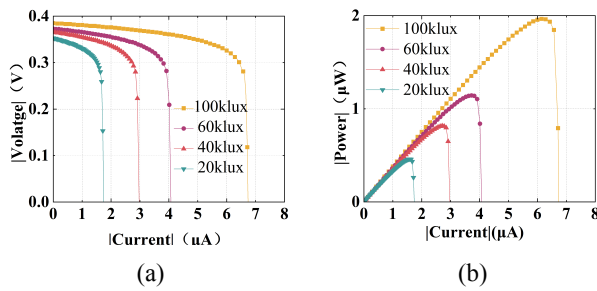


Fig. 4. The measurement results of the EH characteristics for the proposed image sensor at 20klux, 40klux, 60klux and 100klux, respectively, (a) current-voltage curves, (b) current-power curves.

are processed separately. Each quantizer, composed of a ramp generator and a comparator, converts the analog pixel voltages into PWM signals that drive the enable inputs of an 8-bit counter to produce the final digital output [10].

III. EXPERIMENTAL RESULTS

The proposed image sensor was fabricated using a standard $0.18\mu\text{m}$ CMOS process. Fig.3 presents the microscope photograph of the chip, where the dual-channel PWM quantizer adopts a symmetrical design. The counter outputs are buffered to enhance driving capability. The pixel array occupies an area of 0.326 mm^2 . Under a 35.9 fps testing condition, the measured power consumption reveals that the controller consumes $3.98\mu\text{W}$. The analog section, including the pixel array and dual-channel PWM quantizer, consumes $0.292\mu\text{W}$, while the digital section—comprising the controller and counter—consumes $3.688\mu\text{W}$.

To evaluate the energy harvesting capability, we employed Xinyi Optoelectronics' LED chips as the light source, which emit white light with a color temperature of 6500K. Fig. 4(a) presents the load curves under illuminance levels of 20k, 40k, 60k, and 100k lux. The results demonstrate that at 100k lux, the maximum open-circuit voltage reaches $\sim 385\text{ mV}$, while the maximum short-circuit current measures $7\mu\text{A}$. Fig.4(b) shows the current–power characteristic curves of the device. Within a frame period, the measured maximum harvested powers under the four illumination conditions are $0.46\mu\text{W}$, $0.81\mu\text{W}$, $1.14\mu\text{W}$, and $1.96\mu\text{W}$, respectively. Based on these measurement results, the normalized energy-harvesting capability of the device is calculated to be $60.1\text{ pW}/(\text{lux}\cdot\text{mm}^2)$. Furthermore, considering a power management efficiency of approximately 60% as reported in Ref. [12], the proposed image sensor can provide about $1.176\mu\text{W}$ of usable power under 100 klux illumination. Assuming that the sensor power consumption scales approximately linearly with the frame rate and that the harvested power scales approximately linearly with the illumination level, each imaging frame consumes approximately $0.11\mu\text{J}$ of energy. Therefore, the system can theoretically support self-powered operation at approximately 10.7 fps under 100 klux illumination. Similarly, based on the measured relationship between harvested power and illumination, it can be estimated that at an illumination of approximately 9.35 klux, the sensor can support self-powered operation at about 1 frame per second.

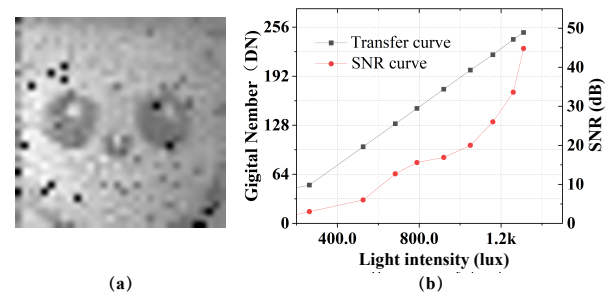


Fig. 5. Performance characterization of the proposed image sensor. (a) Captured images. (b) Photon transfer curve and SNR curve showing the relationship between incident light intensity.

To evaluate the imaging performance of the proposed sensor, The ambient illumination was adjusted to approximately 900 lux using a fill light, and a cartoon image printed on A4 paper was captured. The resulting image, shown in Fig.5(a), demonstrates that the sensor possesses basic imaging functionality. Fig. 5(b) shows the photon transfer curve (PTC) of the sensor. The curve reveals elevated noise levels in low-light regions, consistent with the minor variations observed in the captured sample image, and is likely caused by transistor switching in P-well₂ affecting the photodiode (PD). The sensor's maximum output digital code is 255, and the minimum detectable signal—defined as the smallest standard deviation of the output code—is measured at 1.44. Based on these values, the dynamic range is calculated to be 45dB [13].

TABLE I

PERFORMANCE SUMMARY AND COMPARISON WITH OTHER WORKS

	TCAS-II [5]	TCAS-I [7]	SEN J [9]	This work
Process[μm]	0.18	0.13	0.18	0.18
Pixel type	APS	APS	TFS	APS
Pixel size [μm^2]	393.75	—	—	237.33
Pixel array	32×32	128×96	128×128	32×32
FF[%]	—	44	30.65	55.76
Supply[V]	1.8	1.2	0.7	1
FoM [$\text{pJ}/\text{pixel}\cdot\text{fps}$]	84.7	1131	—	108
DR [dB]	—	—	100	45
P. generation [$\text{pW}/\text{lux}/\mu\text{m}^2$]	31.25	10.55	—	60.1
Intra-frame EH-IMG	No	No	Yes	Yes

IV. CONCLUSION

This letter presents an innovative asynchronous energy-harvesting pixel whose key advantage lies in its self-adaptive energy harvesting mechanism—when the photodiode voltage falls below a threshold, it automatically switches to energy harvesting mode without requiring dedicated control timing, thereby maximizing the energy harvesting window and significantly increasing energy collection time compared to conventional time-division multiplexed pixels. Silicon validation demonstrates that the chip consumes $3.98\mu\text{W}$ at 35.9 fps while maintaining normal imaging functionality. Its energy harvesting capability reaches $1.96\mu\text{W}$ at 100 klux, enabling self-powered operation for 10.7 frames, offering a viable solution for self-powered image sensors.

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